

MODULE 3: PROGRAMMING FOR BIOLOGICAL DATA

Session 6: ANOVA and Linear Regression

Comparing Multiple Groups & Building Calibration Curves

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Learning Objectives



ANOVA

Compare means
across multiple
groups using
Analysis of
Variance.



Post-hoc

Identify exactly
which groups differ
using Tukey HSD.



Regression

Build linear
regression models
for standard
calibration curves.



Validation

Assess model fit
using R^2 and
rigorous residual
analysis.

Clinical Context

Method Applications

Method	Application
ANOVA	Compare 3+ analyzers, reagent lots, or operators.
Regression	Build calibration curves and verify assay linearity.

Today's Scenarios

- **Scenario 1:** Are our three chemistry analyzers giving consistent results?
- **Scenario 2:** Is our assay linear across the reportable range according to CLSI EP06?

Part 1: Analysis of Variance (ANOVA)

Comparing Means Across Multiple Groups

When to Use ANOVA

The Right Test for the Job

Scenario	Statistical Test
Compare 2 groups	T-test
Compare 3+ groups	ANOVA

Warning

Running multiple t-tests inflates the false positive rate!

- Comparing 3 groups requires 3 separate t-tests.
- This leads to a **~14% chance** of finding a false difference.
- ANOVA controls this error by testing all at once.

Scenario: Analyzer Comparison

The Setup

Your lab has **three chemistry analyzers**: A, B, and C.

You run the same glucose QC material 10 times on each instrument.

Question: Are all three analyzers calibrated correctly?

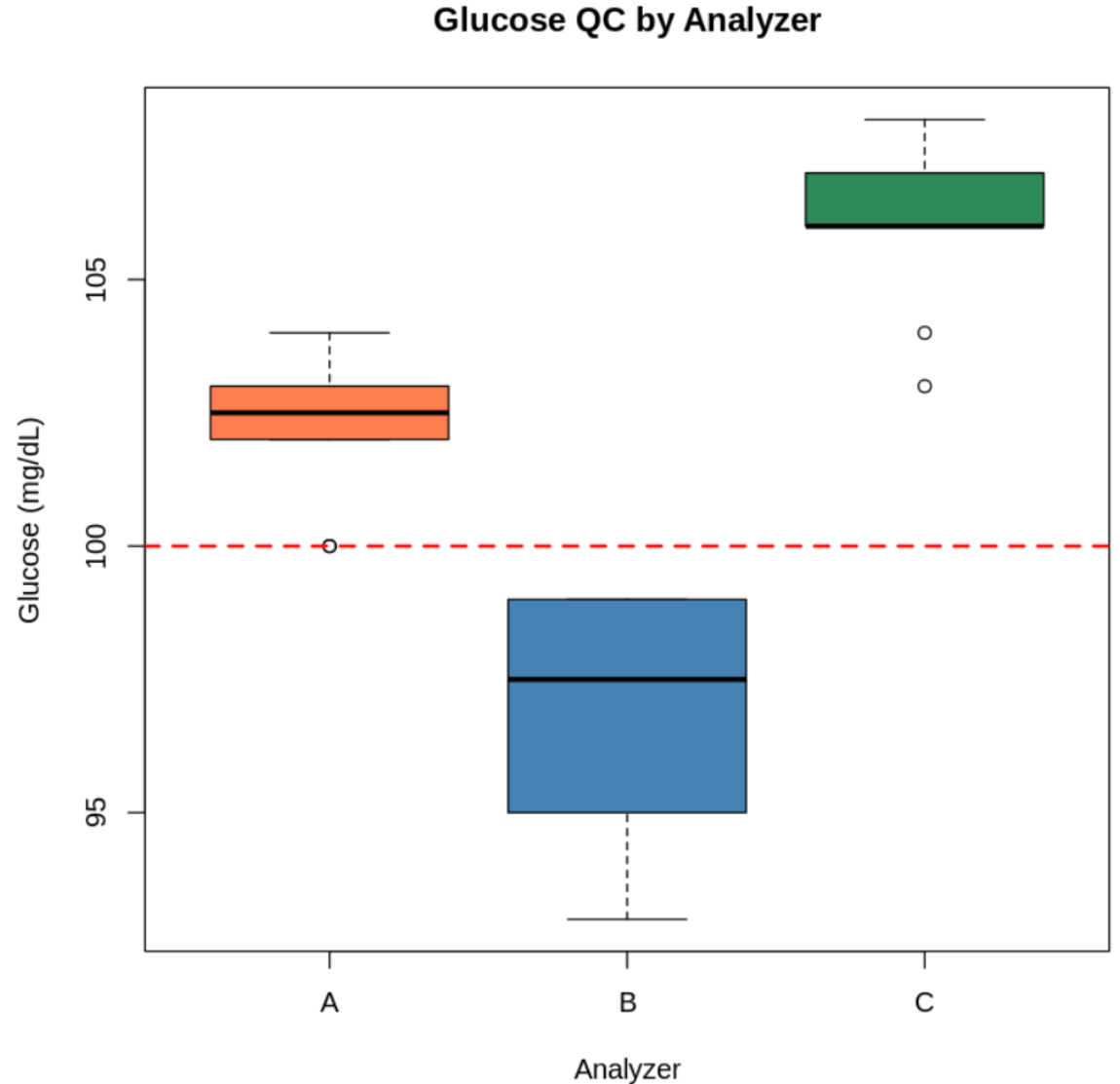
Preliminary Data

Analyzer	Mean Glucose	SD
A	102.3	1.4
B	96.9	2.1
C	106.1	1.6

Visualizing with Boxplots

```
boxplot(Glucose ~ Analyzer,  
        data = analyzer_data,  
        col = c("coral",  
                 "steelblue",  
                 "seagreen"))  
  
# Add target line  
abline(h = 100, col = "red", lty = 2)
```

Visual inspection: The boxplot clearly suggests that the analyzers are producing different results, with C being notably higher.



How ANOVA Works

The F-Statistic

ANOVA compares **between-group variance** (differences between analyzers) to **within-group variance** (precision error).

$$F = \frac{\text{Variance Between Groups}}{\text{Variance Within Groups}}$$

Large F = Groups are different
Small F = Groups are similar

ANOVA Hypotheses

Null Hypothesis (H_0)

All group means are equal.

$$\mu_A = \mu_B = \mu_C$$

Alternative Hypothesis (H_1)

At least one mean differs from the others.

Decision Rule: If $p < 0.05$, we reject H_0 .

ANOVA in R

R Code

```
# Fit ANOVA model
anova_result <- aov(Glucose ~ Analyzer,
                    data = analyzer_data)

# View results
summary(anova_result)
```

Output

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Analyzer	2	427.5	213.73	72.22	1.45e-11 ***
Residuals	27	79.9	2.96		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Interpreting the ANOVA Table

Column	Meaning
Df	Degrees of freedom (Groups - 1)
Sum Sq	Sum of squares (total variance explained)
Mean Sq	Mean square = Sum Sq / Df
F value	Test statistic (Variance Ratio)
Pr(>F)	P-value (The probability of this result if H_0 is true)

$p < 0.05 \rightarrow$ At least one analyzer differs significantly.

The Problem

"ANOVA tells us that 'something' is different, but it doesn't tell us 'what' is different."

*To find the specific culprit, we need a **post-hoc test**.*

Tukey's Honest Significant Difference

Purpose

Perform pairwise comparisons after a significant ANOVA to identify exactly which groups differ.

It controls for the increased error rate of multiple testing.

R Code

```
# Tukey HSD test  
TukeyHSD(anova_result)
```

This compares:

- A vs B
- A vs C
- B vs C

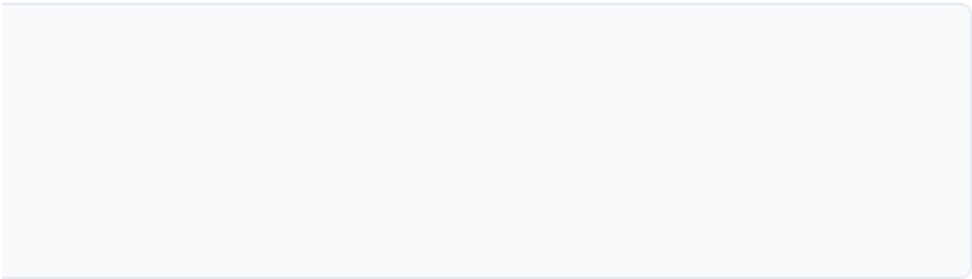
Reading Tukey Output

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = Glucose ~ Analyzer, data = analyzer_data)

\$Analyzer

	diff	lwr	upr	p adj
B-A	-5.4	-7.307463	-3.492537	0.0000004
C-A	3.8	1.892537	5.707463	0.0001034
C-B	9.2	7.292537	11.107463	0.0000000



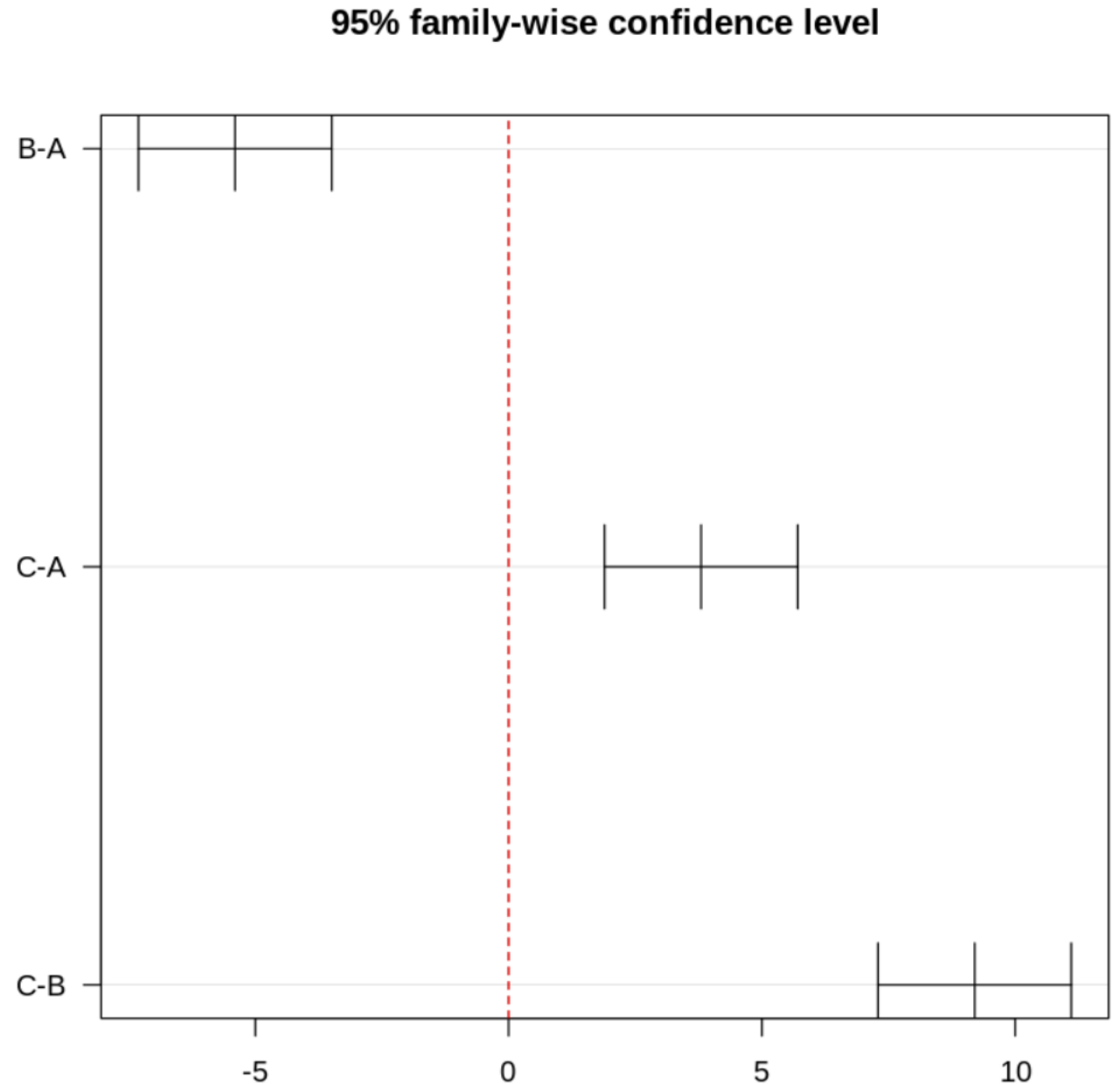
Column	Meaning
diff	Difference between group means.
lwr, upr	95% Confidence Interval of the difference.
p adj	Adjusted p-value.

Tukey Plot

```
plot(TukeyHSD(anova_result))  
abline(v = 0, col = "red", lty = 2)
```

Interpretation:

- The lines represent Confidence Intervals.
- If an interval **crosses zero**, the difference is NOT significant.
- Here, all intervals are far from zero, indicating significant differences for all pairs.



Clinical Action Items

Based on the statistical findings, we take action:

Comparison	Difference	Significant?	Action
B vs A	-7.3	Yes ⚠️	Recalibrate B (reading low)
C vs A	+1.9	Yes ⚠️	Recalibrate C (reading high)
C vs B	+7.3	Yes ⚠️	Major discrepancy detected

Checking ANOVA Assumptions

1

Normality

Residuals should be normally distributed.

```
shapiro.test(residuals(model))
```

2

Homogeneity

Variances should be equal across groups.

```
bartlett.test(Glucose ~ Analyzer)
```

3

Independence

Observations must be independent of each other.

Part 2: Linear Regression & Calibration

Building the Standard Curve

The Foundation of Quantitative Testing

The Concept

Every quantitative assay uses a **calibration curve** to convert measured signals into concentrations.

We model this relationship using the linear equation.

$$y = mx + c$$

Signal = Slope × Concentration + Intercept

CLSI EP06: Linearity Verification

Guideline Requirements

- ✓ Test samples across the full **reportable range**.
- ✓ Verify the **linear relationship** between signal and concentration.
- ✓ Document **recovery** at each level.
- ✓ Identify any **saturation or non-linearity**.



Standard for Lab Compliance

Linearity Verification Data

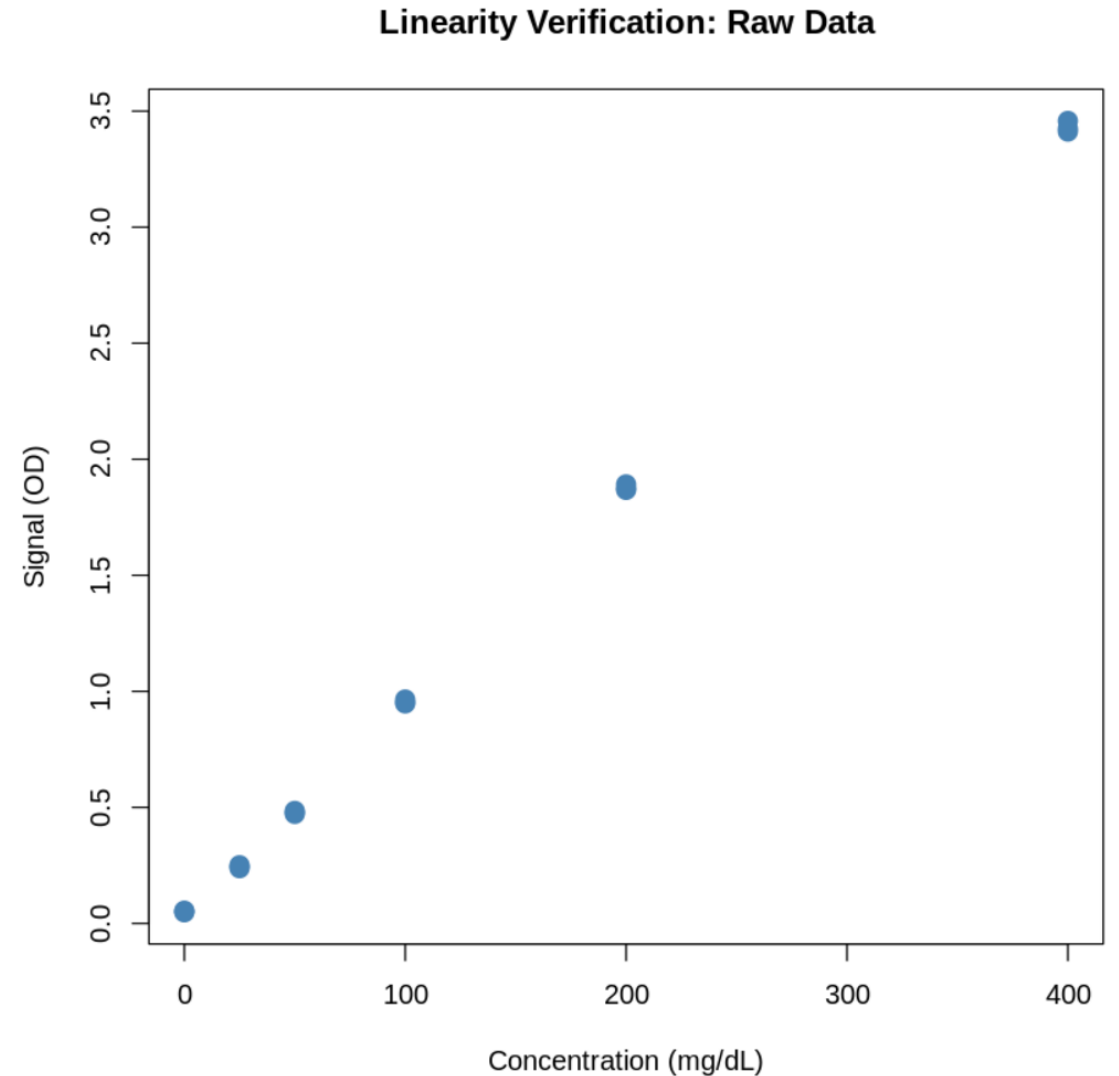
Concentration (mg/dL)	Signal (OD)	Replicates
0 (blank)	0.052	3
25	0.245	3
50	0.478	3
100	0.952	3
200	1.875	3
400	3.425	3

Visualizing Calibration Data

```
plot(Concentration, Signal,  
     main = "Linearity Verification",  
     pch = 19,  
     col = "steelblue")
```

Before fitting a model, always perform a visual check.

Does the relationship look linear?



Linear Regression in R

Fitting the Model

We use the `lm()` function (linear model).

```
model <- lm(Signal ~ Concentration,  
            data = linearity_data)
```

Viewing Results

The `summary()` function provides coefficients and fit statistics.

```
summary(model)
```

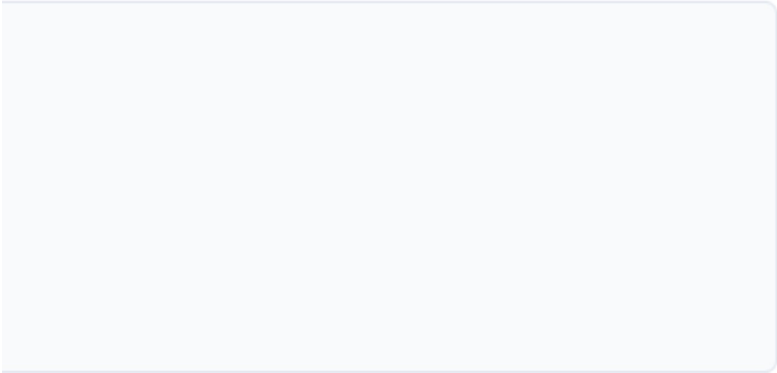
```
Call:
lm(formula = Signal ~ Concentration, data = linearity_data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.07271 -0.03163 -0.01927  0.02664  0.11420
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.090e-02  1.815e-02   3.908  0.00125 **
Concentration  8.535e-03  9.628e-05  88.643 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.05606 on 16 degrees of freedom
Multiple R-squared:  0.998,    Adjusted R-squared:  0.9978
F-statistic: 7858 on 1 and 16 DF,  p-value: < 2.2e-16
```

Interpreting lm() Output



Term	Value	Meaning
Intercept	0.07	The background signal when concentration is 0.
Slope	0.008535	Signal increase per unit of concentration.
R ²	0.998	99.8% of the variance is explained by the model.

Getting the Equation

R Code

```
intercept ← coef(model)[1]  
[REDACTED]  
  
slope ← coef(model)[2]  
[REDACTED]  
  
r_sq ← summary(model)$r.squared  
[REDACTED]
```

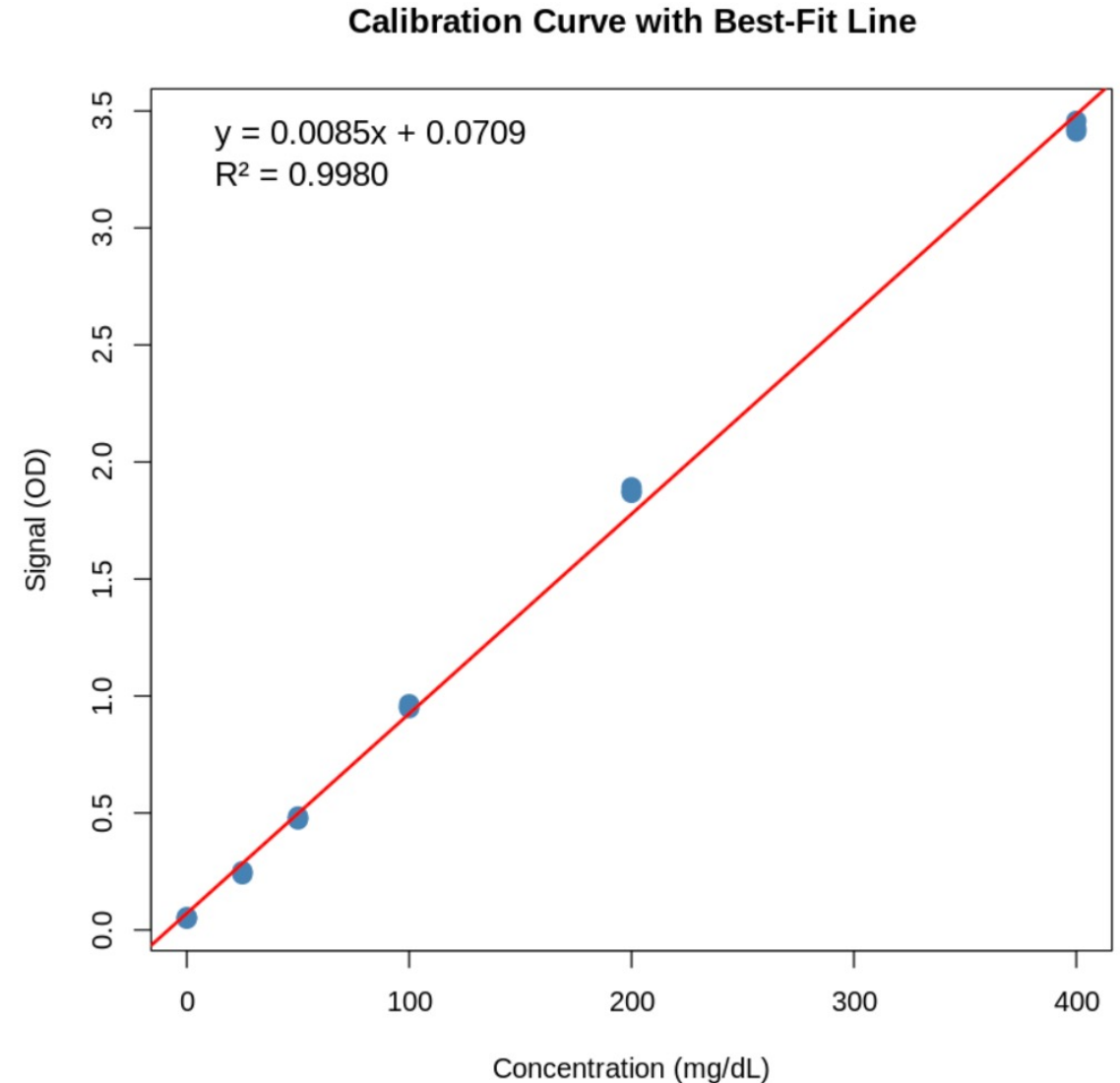
Final Calibration Equation:

Signal = 0.0085 × Concentration + 0.0709

Adding the Regression Line

```
plot(Concentration, Signal,  
     pch = 19, col = "steelblue")  
  
# Add the regression line  
abline(model, col = "red", lwd = 2)
```

This visualizes how well our calculated model fits the actual data points.



Understanding R²

Coefficient of Determination

$$R^2 = 1 - \frac{SS_{\text{residual}}}{SS_{\text{total}}}$$

Interpretation

R ² Value	Meaning
0.99+	Excellent fit (Standard for Clinical Labs)
0.95–0.99	Good fit
< 0.90	Poor fit — Investigate!

Why Check Residuals?

The Concept

Residual = Observed - Predicted

Residuals allow us to see patterns that R^2 might hide.

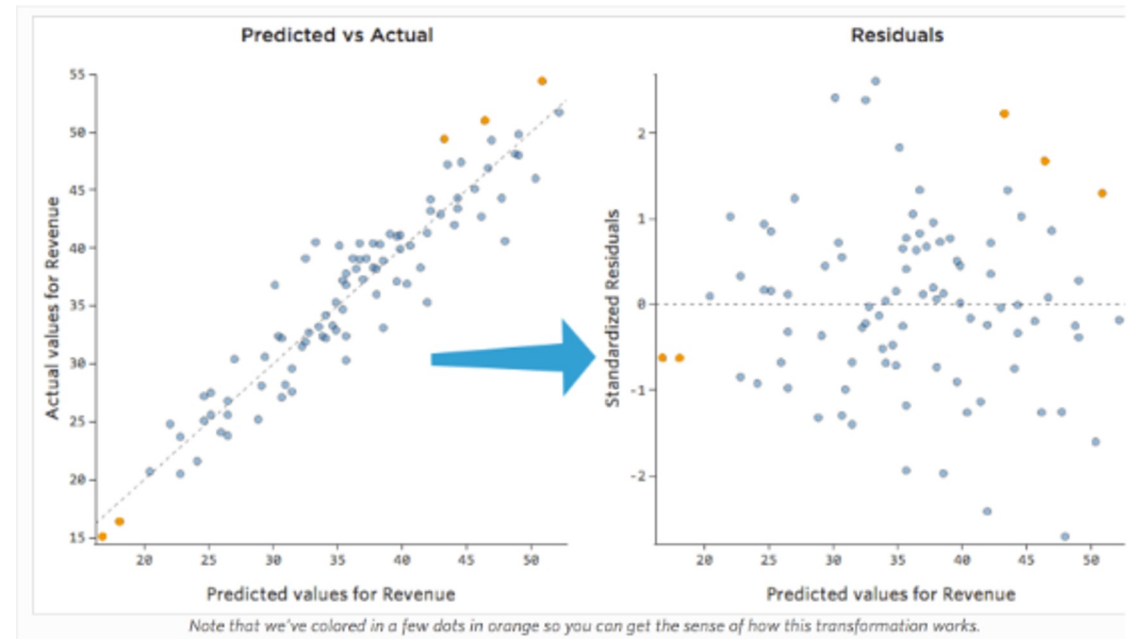
What to Look For

- ✓ **Random Scatter:** Good model.
- ❗ **Curvature:** Non-linearity (needs polynomial fit).
- ❗ **Funnel Shape:** Heteroscedasticity (unequal variance).

Creating a Residual Plot

```
plot(Concentration, residuals(model),  
     main = "Residual Plot",  
     pch = 19)
```

```
# Add zero line  
abline(h = 0, col = "red", lty = 2)
```



Built-in Diagnostic Plots

```
par(mfrow = c(2,2))  
plot(model)
```

What R Generates

Plot	Checks For
Residuals vs Fitted	Linearity
Q-Q Plot	Normality of Residuals
Scale-Location	Homoscedasticity
Residuals vs Leverage	Influential Outliers

Using the Calibration Curve

$$\text{Concentration} = \frac{\text{Signal-Intercept}}{\text{Slope}}$$

R Code

```
unknown_signal ← 1.5  
  
predicted_conc ← (unknown_signal -  
intercept) / slope
```

Linearity Acceptance Criteria

Criterion	Typical Requirement
R ²	≥ 0.99
Residuals	Within ±10% of predicted value
Recovery	95–105% at each concentration level

Note: Specific requirements vary by analyte and method SOP.

Automated Verification Check

```
if (r_squared >= 0.99) { cat("✅ R² PASS") } else { cat("⚠️ R² FAIL") } if  
(max(abs(residuals)) <= 0.10 * predicted) { cat("✅ Residuals PASS") } else { cat("⚠️  
Residuals FAIL") }
```

Summary: ANOVA vs. Regression

Feature	ANOVA	Linear Regression
Purpose	Compare group means	Model relationships
Question	"Are groups different?"	"What is the equation?"
R Function	aov()	lm()
Key Statistics	F-value, p-value	R ² , slope, intercept
Lab Use	Analyzer/Lot Comparison	Calibration Curves

Key Takeaways

ANOVA

Use for comparing 3+ groups.
Always follow up with Tukey
HSD.

Regression

Models $y = mx + c$. Use $R^2 \geq 0.99$ as a quality standard.

Residuals

Essential for validating that
your model assumptions are
met.

Tutorial 6

Calibration Verification in Practice

Open your R environment

Tutorial 6: Calibration Verification

Building and Validating Standard Curves

Your Scenario

The Task

You are performing **annual linearity verification** following CLSI EP06 guidelines.

- Verify linear response across reportable range.
- Document calibration equation.
- Ensure $R^2 \geq 0.99$.

The Data

6 Concentration Levels

Each measured in **triplicate**.

What You'll Practice

Step	Skill	Time
1-2	Load and explore data	10 min
3	Fit linear model	15 min
4	Create calibration plot	15 min
5	Analyze residuals	15 min
6-7	Calculate unknowns & Report	15 min

Steps 1-2: Load and Visualize

Load Data

```
linearity_data <- data.frame(  
  Concentration = rep(c(0, 25, 50, 100, 200,  
400), each = 3),  
  Signal = c(0.052, 0.048, ... )  
)
```

Visualize

```
str(linearity_data)  
  
aggregate(Signal ~ Conc, FUN = mean)  
  
plot(Concentration, Signal)
```


Step 3: Fit Linear Model

The Magic Line

```
model <- lm(Signal ~ Concentration,  
            data = linearity_data)  
  
summary(model)
```

Extract Parameters

```
intercept <- coef(model)[1]  
slope <- coef(model)[2]  
r_squared <- summary(model)$r.squared
```

Step 4: Calibration Plot

Base R

```
plot(Concentration, Signal, pch = 19)  
abline(model, col = "red", lwd = 2)
```

ggplot2 (Publication Quality)

```
ggplot(data, aes(x=Concentration,  
y=Signal)) +  
  geom_point(size = 3) +  
  geom_smooth(method = "lm", se = TRUE)
```

Step 5: Check Residuals

Why?

Residuals reveal problems R^2 misses:

- Curvature = Non-linearity
- Funnel = Unequal variance

Code

```
plot(Concentration, residuals(model))  
abline(h = 0, col = "red", lty = 2)
```

Goal: Random scatter around zero.

Step 6: Predict Unknowns

$$\text{Conc} = \frac{\text{Signal-Intercept}}{\text{Slope}}$$

Calculation

```
unknown_signal ← 1.534  
predicted_conc ← (unknown_signal -  
intercept) / slope
```

Step 7: Verification Report

Checklist

- Date and Analyst
- Calibration Equation
- R^2 Value (Pass/Fail)
- Conclusion

Example Conclusion

"Linearity verification PASSED. $R^2 = 0.9987$ exceeds acceptance criterion of 0.99. Assay is acceptable for clinical use."

Questions?

Session 6 Complete